Geophysical Models and Numerical Standards in ETideLoad4.5

ETideLoad4.5; https://www.zcyphygeodesy.com/en/

ZHANG Chuanyin; zpmzsyzcy1968@163.com

Chinese Academy of Surveying & Mapping

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ETideLoad4.5 is primarily founded upon the geophysical models and numerical standards recommended by the IERS Conventions (2010). These models and standards are configurable via the utility [System Configs for the Geophysical Models and Numerical Standards]. All associated data files are archived under the directory: C:\ETideLoad4.5_win64en\iers.

7.4.1 The Surface Atmosphere Tidal Load Spherical Harmonic Coefficient Model File

The 360-degree surface atmosphere tidal load spherical harmonic coefficient model, ECMWF2006.dat, resides in the iers folder and adheres to the FES2004 format. This model was constructed using the spherical harmonic analysis modules of ETideLoad4.5, based on 0.5° × 0.5° global harmonic constant grids derived from the ECMWF-DCDA2006 reanalysis data (European Centre for Medium-Range Weather Forecasts). It encompasses four principal atmospheric tidal constituents – diurnal (S_1), semi-diurnal (S_2), semi-annual (Ssa), and annual (Sa) – to satisfy the precision requirements of centimeter-level geodesy.

ECMWF2006.dat

```
Atmospheric tide model: ECMWF-DCDA2006 normalized model up to (360,360) in hPa
半日/周日/半年/年周期
Doodson   Darw   n    m       Csin+         Ccos+         Csin-         Ccos-          C+         eps+          C-         eps-
 164.556   S1    1    0   -0.01055351    0.00555959   -0.01055351    0.00555959   0.01192835  297.7803   0.01192835  297.7803
 164.556   S1    2    0   -0.00898730    0.02713172   -0.00898730    0.02713172   0.02858149  341.6727   0.02858149  341.6727
 164.556   S1    3    0    0.02416514    0.01232573    0.02416514    0.01232573   0.02712707   62.9756   0.02712707   62.9756
 164.556   S1    4    0    0.01971779   -0.01808456    0.01971779   -0.01808456   0.02675523  132.5261   0.02675523  132.5261
 164.556   S1    5    0    0.00538826   -0.01556217    0.00538826   -0.01556217   0.01646859  160.9021   0.01646859  160.9021
 164.556   S1    6    0   -0.01896560   -0.00055330   -0.01896560   -0.00055330   0.01897366  268.3289   0.01897366  268.3289
 164.556   S1    7    0    0.00163224    0.00711629    0.00163224    0.00711629   0.00730108   12.9183   0.00730108   12.9183
 164.556   S1    8    0    0.00341644    0.00607435    0.00341644    0.00607435   0.00696920   29.3550   0.00696920   29.3550
 164.556   S1    9    0   -0.00469730   -0.00311697   -0.00469730   -0.00311697   0.00563739  236.4331   0.00563739  236.4331
 164.556   S1   10    0    0.00442735   -0.01563001    0.00442735   -0.01563001   0.01624496  164.1847   0.01624496  164.1847
 164.556   S1   11    0    0.00941838   -0.00082619    0.00941838   -0.00082619   0.00945455   95.0132   0.00945455   95.0132
 164.556   S1   12    0   -0.00454013    0.00688423   -0.00454013    0.00688423   0.00824654  326.5953   0.00824654  326.5953
 164.556   S1   13    0   -0.01227672    0.00310149   -0.01227672    0.00310149   0.01266243  284.1781   0.01266243  284.1781
 164.556   S1   14    0    0.00203678    0.00166923    0.00203678    0.00166923   0.00263340   50.6638   0.00263340   50.6638
 164.556   S1   15    0    0.00253994    0.00381849    0.00253994    0.00381849   0.00458608   33.6306   0.00458608   33.6306
 164.556   S1   16    0    0.00613602   -0.00041704    0.00613602   -0.00041704   0.00615017   93.8882   0.00615017   93.8882
 164.556   S1   17    0   -0.00113104   -0.00413462   -0.00113104   -0.00413462   0.00428652  195.2992   0.00428652  195.2992
 164.556   S1   18    0   -0.00311700    0.00136741   -0.00311700    0.00136741   0.00340375  293.6868   0.00340375  293.6868
 164.556   S1   19    0   -0.00217138    0.00053937   -0.00217138    0.00053937   0.00223737  283.9498   0.00223737  283.9498
 164.556   S1   20    0   -0.00017645    0.00369644   -0.00017645    0.00369644   0.00370065  357.2671   0.00370065  357.2671
 164.556   S1   21    0    0.00068441   -0.00165216    0.00068441   -0.00165216   0.00178831  157.4980   0.00178831  157.4980
 164.556   S1   22    0    0.00100221   -0.00214635    0.00100221   -0.00214635   0.00236881  154.9703   0.00236881  154.9703
 164.556   S1   23    0    0.00461395   -0.00179653    0.00461395   -0.00179653   0.00495136  111.2744   0.00495136  111.2744
 164.556   S1   24    0   -0.00143873    0.00014453   -0.00143873    0.00014453   0.00144597  275.7366   0.00144597  275.7366
```

Note: Surface atmospheric pressure, tidal constituent harmonic constants, and tidal load spherical harmonic coefficients are uniformly expressed in hPa (or mbar).

7.4.2 The Ocean Tidal Load Spherical Harmonic Coefficient Model File

FES2004S1.dat: The 100-degree ocean tidal load spherical harmonic coefficient model is stored in the iers folder in FES2004 format. The relationship between the normalized

ocean tidal load spherical harmonic coefficients and the geopotential coefficients follows Equation (6.15) of the IERS Conventions (2010).

FES2014b720cs.dat: To accommodate high-precision processing for satellite altimetry, coastal zones, and ocean gravity gradiometry, the AVISO FES2014b global tidal height harmonic constant grid models were utilized. The resulting 720-degree ocean tidal height spherical harmonic coefficient model is stored in FES2004 format. This file includes coefficients for 36 tidal constituents (Ω_1, Ω_2; $2N_2$, ε_2, J_1, K_1, K_2, L_2, λ_2, M_2, M_3, M_4, M_6, M_8, Mf, MKS_2, Mm, MN_4, MS_4, MSf, MSqm, Mtm, μ_2, N_2, N_4, ν_2, O_1, P_1, Q_1, R_2, S_1, S_2, S_4, Sa, Ssa, T_2). The coefficients for the two equilibrium tidal constituent (Ω_1, Ω_2) are inherited from FES2004S1.dat.

BCMWF2006.dat | FES2004S1.dat

```
Ocean tide model: FES2004 normalized model (fev. 2004) up to (100,100) in cm
(long period from FES2002 up to (50,50) + equilibrium Om1/Om2, atmospheric tide NOT included)
Doodson Darw    n   m     Csin+      Ccos+       Csin-      Ccos-      C+     eps+      C-      eps-
  55.565 Om1    2   0 -0.540594  0.000000    0.000000   0.000000   0.5406 270.000 0.0000   0.000
  55.575 Om2    2   0 -0.005218  0.000000    0.000000   0.000000   0.0052 270.000 0.0000   0.000
  56.554 Sa     1   0  0.017233  0.000013    0.000000   0.000000   0.0172  89.957 0.0000   0.000
  56.554 Sa     2   0 -0.046604 -0.000903    0.000000   0.000000   0.0466 268.890 0.0000   0.000
  56.554 Sa     3   0 -0.000889  0.000049    0.000000   0.000000   0.0009 273.155 0.0000   0.000
  56.554 Sa     4   0  0.012069 -0.000413    0.000000   0.000000   0.0121  91.960 0.0000   0.000
  56.554 Sa     5   0 -0.009780 -0.000421    0.000000   0.000000   0.0098 267.535 0.0000   0.000
  56.554 Sa     6   0  0.006895  0.000043    0.000000   0.000000   0.0069  89.643 0.0000   0.000
  56.554 Sa     7   0 -0.010515 -0.000287    0.000000   0.000000   0.0105 268.437 0.0000   0.000
  56.554 Sa     8   0  0.002067 -0.000011    0.000000   0.000000   0.0021  90.305 0.0000   0.000
  56.554 Sa     9   0 -0.004236 -0.000110    0.000000   0.000000   0.0042 268.512 0.0000   0.000
  56.554 Sa    10   0 -0.001781 -0.000085    0.000000   0.000000   0.0018 267.268 0.0000   0.000
  56.554 Sa    11   0 -0.001372 -0.000068    0.000000   0.000000   0.0014 267.163 0.0000   0.000
  56.554 Sa    12   0 -0.004081 -0.000048    0.000000   0.000000   0.0041 269.326 0.0000   0.000
  56.554 Sa    13   0 -0.000116 -0.000041    0.000000   0.000000   0.0001 250.534 0.0000   0.000
  56.554 Sa    14   0 -0.003043 -0.000007    0.000000   0.000000   0.0030 269.868 0.0000   0.000
  56.554 Sa    15   0  0.001109 -0.000028    0.000000   0.000000   0.0011  91.446 0.0000   0.000
  56.554 Sa    16   0 -0.002596 -0.000034    0.000000   0.000000   0.0026 269.250 0.0000   0.000
  56.554 Sa    17   0 -0.000674  0.000022    0.000000   0.000000   0.0007 271.870 0.0000   0.000
  56.554 Sa    18   0  0.000546  0.000006    0.000000   0.000000   0.0005  89.370 0.0000   0.000
  56.554 Sa    19   0 -0.000024  0.000023    0.000000   0.000000   0.0000 313.781 0.0000   0.000
  56.554 Sa    20   0  0.000867  0.000014    0.000000   0.000000   0.0009  89.075 0.0000   0.000
```

Note: Ocean tidal heights, harmonic constants, and tidal load spherical harmonic coefficients are uniformly expressed in cm.

7.4.3 The Earth's Load Love Number File

The Earth's Load Love numbers (LDC), representing load deformation coefficients, are computed based on the spherically symmetric, non-rotating, elastic Earth model PREM (Dziewonski & Anderson, 1981). In ETideLoad4.5, these values are derived from the Regional EIAstic Rebound calculator (REAR 1.0, 2015.11).

Love_load_cm.dat

```
The load Love numbers from the REAR package are attached. There are no
more of these oscillations at high degree, and they go up to degree 32768.
November 20, 2015, Jean-Paul
CM: center of mass reference frame
    n     h' (vert)         l' (horiz)         k' (potent)
    0   0.0000000000D+00  0.0000000000D+00   0.0000000000D+00
    1  -0.0287112988D+01  0.1045044062D+00   0.0000000000D+00
    2  -0.9945870591D+00  0.2411251588D-01  -0.3057703360D+00
    3  -0.1054653021D+01  0.7085493677D-01  -0.1962722363D+00
    4  -0.1057783895D+01  0.5958723183D-01  -0.1337905897D+00
    5  -0.1091185915D+01  0.4702627503D-01  -0.1047617976D+00
    6  -0.1149253656D+01  0.3940811757D-01  -0.9034958051D-01
    7  -0.1218363201D+01  0.3499400649D-01  -0.8205733906D-01
    8  -0.1290473661D+01  0.3225123202D-01  -0.7652348967D-01
    9  -0.1361847865D+01  0.3038562458D-01  -0.7239287690D-01
   10  -0.1430981761D+01  0.2902258995D-01  -0.6907768441D-01
   11  -0.1497377458D+01  0.2798156018D-01  -0.6629382122D-01
   12  -0.1560934855D+01  0.2716367080D-01  -0.6388475059D-01
   13  -0.1621715593D+01  0.2650554043D-01  -0.6175536119D-01
   14  -0.1679770379D+01  0.2596800569D-01  -0.5983856019D-01
   15  -0.1735198310D+01  0.2551661917D-01  -0.5808965155D-01
   16  -0.1788088250D+01  0.2512667367D-01  -0.5647488828D-01
   17  -0.1838448069D+01  0.2478452380D-01  -0.5496610314D-01
   18  -0.1886440474D+01  0.2447083426D-01  -0.5354901315D-01
   19  -0.1932084480D+01  0.2417919471D-01  -0.5220607051D-01
   20  -0.1975465902D+01  0.2389862142D-01  -0.5092726303D-01
   21  -0.2016677975D+01  0.2362510597D-01  -0.4970406011D-01
   22  -0.2055800328D+01  0.2335504487D-01  -0.4853059813D-01
   23  -0.2092911079D+01  0.2308664225D-01  -0.4740132374D-01
   24  -0.2128152865D+01  0.2281672671D-01  -0.4631386954D-01
```

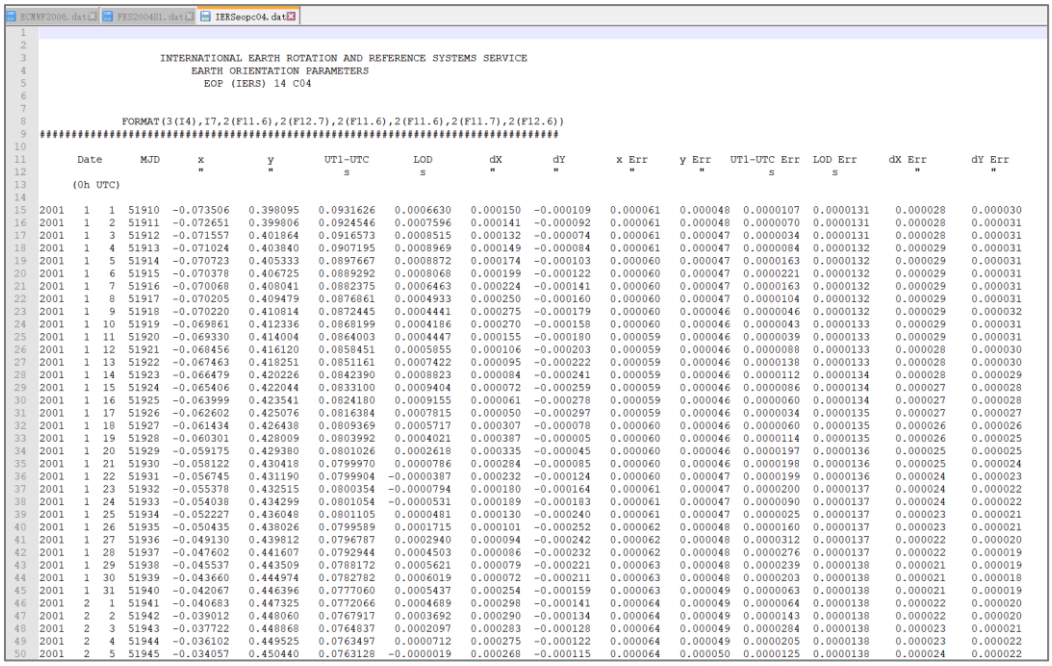
The file Love_load_cm.dat, located in the iers folder, contains the load Love numbers for radial displacement ($h_n^{'}$), horizontal displacement (l_n'), and geopotential (k_n') for degrees $\mathrm{n} = 0,...,32768$, with boundary conditions $k_0' = k_1' = 0$.

To suppress high-degree oscillations in the load Green's function, calculations extend up to degree 54,000. For degrees exceeding the tabulated limit ($\mathrm{n} > 32768$), the following asymptotic formulas are employed:

h_n' =-6.209114, l_n' =1.890061/n, k_n' =-2.682697/n.

7.4.4 The IERS Earth Orientation Parameter Time Series File

The Earth Orientation Parameters (EOP) time series file, IERSeopc04.dat (referenced to ITRF2008), is stored in the iers folder. This dataset can be updated via the IERS website. For future epochs, forecast EOP products may be utilized; however, given the non-tidal nature of polar motion, the forecast horizon should be restricted to within six months.



EGM96.dat EIGEN2006.dat FES2004SL.dat IERSeopc04.dat

```
                    INTERNATIONAL EARTH ROTATION AND REFERENCE SYSTEMS SERVICE
                        EARTH ORIENTATION PARAMETERS
                         EOP (IERS) 14 C04

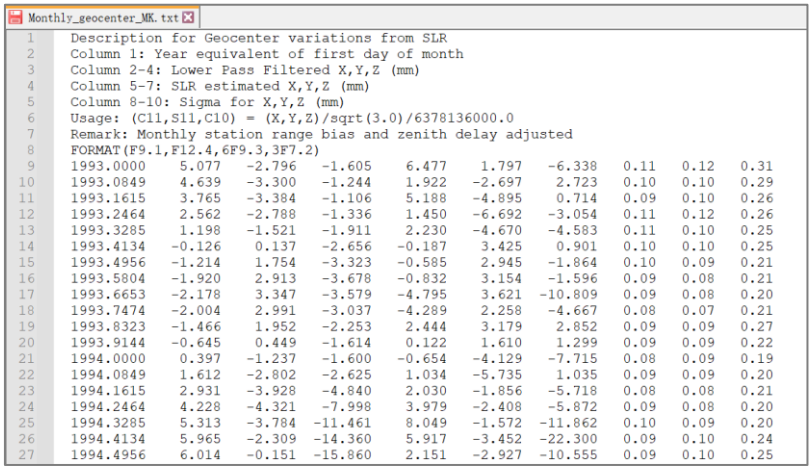

                FORMAT(3(I4),I7,2(F11.6),2(F12.7),2(F11.6),2(F11.6),2(F11.7),2(F12.6))
#########################################################################################

     Date      MJD        x          y         UT1-UTC      LOD         dX         dY        x Err      y Err   UT1-UTC Err  LOD Err    dX Err     dY Err
                          "          "            s          s          "          "          "          "          s          s          "          "
   (0h UTC)

2001  1  1  51910  -0.073506   0.398095   0.0931626   0.0006630   0.000150  -0.000109   0.000061   0.000048   0.0000107  0.0000131  0.000028   0.000030
2001  1  2  51911  -0.072651   0.399806   0.0924546   0.0007596   0.000141  -0.000092   0.000061   0.000048   0.0000070  0.0000131  0.000028   0.000031
2001  1  3  51912  -0.071557   0.401864   0.0916573   0.0008515   0.000132  -0.000074   0.000061   0.000047   0.0000034  0.0000131  0.000028   0.000031
2001  1  4  51913  -0.071024   0.403840   0.0907195   0.0008969   0.000149  -0.000084   0.000061   0.000047   0.0000084  0.0000132  0.000029   0.000031
2001  1  5  51914  -0.070723   0.405333   0.0897667   0.0008872   0.000174  -0.000103   0.000060   0.000047   0.0000163  0.0000132  0.000029   0.000031
2001  1  6  51915  -0.070378   0.406725   0.0889292   0.0008068   0.000199  -0.000122   0.000060   0.000047   0.0000221  0.0000132  0.000029   0.000031
2001  1  7  51916  -0.070068   0.408041   0.0882375   0.0006463   0.000224  -0.000141   0.000060   0.000047   0.0000163  0.0000132  0.000029   0.000031
2001  1  8  51917  -0.070205   0.409479   0.0876861   0.0004933   0.000250  -0.000160   0.000060   0.000047   0.0000104  0.0000132  0.000029   0.000031
2001  1  9  51918  -0.070220   0.410814   0.0872445   0.0004441   0.000275  -0.000179   0.000060   0.000046   0.0000046  0.0000132  0.000029   0.000032
2001  1 10  51919  -0.069861   0.412336   0.0868199   0.0004186   0.000270  -0.000158   0.000060   0.000046   0.0000043  0.0000133  0.000029   0.000031
2001  1 11  51920  -0.069330   0.414004   0.0864063   0.0004447   0.000155  -0.000180   0.000059   0.000046   0.0000039  0.0000133  0.000029   0.000031
2001  1 12  51921  -0.068456   0.416120   0.0858451   0.0005855   0.000106  -0.000203   0.000059   0.000046   0.0000088  0.0000133  0.000028   0.000030
2001  1 13  51922  -0.067463   0.418251   0.0851161   0.0007422   0.000095  -0.000222   0.000059   0.000046   0.0000138  0.0000133  0.000028   0.000030
2001  1 14  51923  -0.066479   0.420226   0.0842390   0.0008823   0.000084  -0.000241   0.000059   0.000046   0.0000112  0.0000134  0.000028   0.000029
2001  1 15  51924  -0.065406   0.422044   0.0833100   0.0009404   0.000072  -0.000259   0.000059   0.000046   0.0000086  0.0000134  0.000027   0.000028
2001  1 16  51925  -0.063999   0.423541   0.0824180   0.0009155   0.000061  -0.000278   0.000059   0.000046   0.0000060  0.0000134  0.000027   0.000028
2001  1 17  51926  -0.062602   0.425076   0.0816384   0.0007815   0.000050  -0.000297   0.000059   0.000046   0.0000034  0.0000135  0.000027   0.000027
2001  1 18  51927  -0.061434   0.426438   0.0809369   0.0005717   0.000307  -0.000078   0.000060   0.000046   0.0000060  0.0000135  0.000026   0.000026
2001  1 19  51928  -0.060301   0.428009   0.0803992   0.0004021   0.000387  -0.000005   0.000060   0.000046   0.0000114  0.0000135  0.000026   0.000025
2001  1 20  51929  -0.059175   0.429380   0.0801026   0.0002618   0.000335  -0.000045   0.000060   0.000046   0.0000197  0.0000136  0.000025   0.000025
2001  1 21  51930  -0.058122   0.430418   0.0799770   0.0000786   0.000284  -0.000085   0.000060   0.000046   0.0000198  0.0000136  0.000025   0.000024
2001  1 22  51931  -0.056745   0.431190   0.0799904  -0.0000387   0.000232  -0.000124   0.000060   0.000047   0.0000199  0.0000136  0.000024   0.000023
2001  1 23  51932  -0.055378   0.432515   0.0800354  -0.0000794   0.000180  -0.000164   0.000061   0.000047   0.0000200  0.0000137  0.000024   0.000022
2001  1 24  51933  -0.054038   0.434299   0.0801054  -0.0000531   0.000189  -0.000183   0.000061   0.000047   0.0000090  0.0000137  0.000024   0.000022
2001  1 25  51934  -0.052227   0.436048   0.0801105   0.0000481   0.000130  -0.000240   0.000061   0.000047   0.0000025  0.0000137  0.000023   0.000021
2001  1 26  51935  -0.050435   0.438026   0.0799589   0.0001715   0.000101  -0.000252   0.000062   0.000048   0.0000160  0.0000137  0.000023   0.000021
2001  1 27  51936  -0.049130   0.439812   0.0796787   0.0002940   0.000094  -0.000242   0.000062   0.000048   0.0000312  0.0000137  0.000022   0.000020
2001  1 28  51937  -0.047602   0.441607   0.0792944   0.0004503   0.000086  -0.000232   0.000062   0.000048   0.0000276  0.0000137  0.000022   0.000019
2001  1 29  51938  -0.045537   0.443509   0.0788172   0.0005621   0.000079  -0.000221   0.000063   0.000048   0.0000239  0.0000138  0.000021   0.000019
2001  1 30  51939  -0.043660   0.444974   0.0782782   0.0006019   0.000072  -0.000211   0.000063   0.000048   0.0000203  0.0000138  0.000021   0.000018
2001  1 31  51940  -0.042067   0.446396   0.0777060   0.0005437   0.000254  -0.000159   0.000063   0.000049   0.0000063  0.0000138  0.000021   0.000019
2001  2  1  51941  -0.040683   0.447325   0.0772066   0.0004689   0.000298  -0.000141   0.000064   0.000049   0.0000064  0.0000138  0.000022   0.000020
2001  2  2  51942  -0.039012   0.448060   0.0767917   0.0003692   0.000290  -0.000134   0.000064   0.000049   0.0000143  0.0000138  0.000022   0.000020
2001  2  3  51943  -0.037722   0.448868   0.0764837   0.0002097   0.000283  -0.000128   0.000064   0.000049   0.0000284  0.0000138  0.000023   0.000021
2001  2  4  51944  -0.036102   0.449525   0.0763497   0.0000712   0.000275  -0.000122   0.000064   0.000049   0.0000205  0.0000138  0.000023   0.000022
2001  2  5  51945  -0.034057   0.450440   0.0763128  -0.0000019   0.000268  -0.000115   0.000064   0.000050   0.0000125  0.0000138  0.000024   0.000022
```

7.4.5 The Earth's COM Variation Time Series File

The Earth's Center of Mass (COM) variation time series, Monthly_geocenter_MK.txt (referenced to ITRF2014), is provided by the Center for Space Research at the University of Texas (UT/CSR). These data are derived from Satellite Laser Ranging (SLR) observations of LAGEOS-1/2, Stella, Starlette, AJISAI, BEC, and LARES satellites. Forecast products may be used for future epochs, limited to a duration of three months.



Monthly_geocenter_MK.txt

```
Description for Geocenter variations from SLR
Column 1: Year equivalent of first day of month
Column 2-4: Lower Pass Filtered X,Y,Z (mm)
Column 5-7: SLR estimated X,Y,Z (mm)
Column 8-10: Sigma for X,Y,Z (mm)
Usage: (C11,S11,C10) = (X,Y,Z)/sqrt(3.0)/6378136000.0
Remark: Monthly station range bias and zenith delay adjusted
FORMAT(F9.1,F12.4,6F9.3,3F7.2)
1993.0000    5.077   -2.796   -1.605    6.477    1.797   -6.338   0.11   0.12   0.31
1993.0849    4.639   -3.300   -1.244    1.922   -2.697    2.723   0.10   0.10   0.29
1993.1615    3.765   -3.384   -1.106    5.188   -4.895    0.714   0.09   0.10   0.26
1993.2464    2.562   -2.788   -1.336    1.450   -6.692   -3.054   0.11   0.12   0.26
1993.3285    1.198   -1.521   -1.911    2.230   -4.670   -4.583   0.11   0.10   0.25
1993.4134   -0.126    0.137   -2.656   -0.187    3.425    0.901   0.10   0.10   0.25
1993.4956   -1.214    1.754   -3.323   -0.585    2.945   -1.864   0.10   0.09   0.21
1993.5804   -1.920    2.913   -3.678   -0.832    3.154   -1.596   0.09   0.08   0.21
1993.6653   -2.178    3.347   -3.579   -4.795    3.621  -10.809   0.09   0.08   0.20
1993.7474   -2.004    2.991   -3.037   -4.289    2.258   -4.667   0.08   0.07   0.21
1993.8323   -1.466    1.952   -2.253    2.444    3.179    2.852   0.09   0.09   0.27
1993.9144   -0.645    0.449   -1.614    0.122    1.610    1.299   0.09   0.09   0.22
1994.0000    0.397   -1.237   -1.600   -0.654   -4.129   -7.715   0.08   0.09   0.19
1994.0849    1.612   -2.802   -2.625    1.034   -5.735    1.035   0.09   0.09   0.20
1994.1615    2.931   -3.928   -4.840    2.030   -1.856   -5.718   0.08   0.08   0.21
1994.2464    4.228   -4.321   -7.998    3.979   -2.408   -5.872   0.09   0.08   0.20
1994.3285    5.313   -3.784  -11.461    8.049   -1.572  -11.862   0.10   0.09   0.20
1994.4134    5.965   -2.309  -14.360    5.917   -3.452  -22.300   0.09   0.10   0.24
1994.4956    6.014   -0.151  -15.860    2.151   -2.927  -10.555   0.09   0.10   0.25
```

7.4.6 Ocean Tidal Constituent Harmonic Constant Grid Model Files

Composition: An ocean tidal height model comprises multiple vector grid files, each storing the harmonic constants for a specific tidal constituent.

Directory Structure: All grid files belonging to a single ocean tidal height model must

reside in the same directory and share identical grid specifications (extent and resolution).

Example: The directory C:\ETideLoad4.5_win64en\OceanTide contains 10 vector grid files representing the GOT4.8 ocean tide model.

Identification: The specific tidal constituent is identified by the seventh attribute (Doodson number) in the grid file header. Filenames are user-definable.

Units: Models may be global or regional. All ocean tidal heights and harmonic constants are expressed in cm.

M2_got4.8.dat

```
0.000000  360.000000  -90.000000  90.000000  0.50000000  0.50000000  255555
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
```

7.4.7 The JPL Planetary Ephemeris DE440 File

The JPL Planetary Ephemeris file JEPH.440 (covering the epoch range 1850 – 2201) is stored in the iers folder.

7.4.8 The Love Number Correction File for Frequency Dependence

The file frqadjlovekhl.txt contains corrections for the frequency dependence of Love numbers. Generated from Tables 6.5a – c and 7.2 – 7.3b of the IERS Conventions (2010), this file is essential for calculating high-accuracy solid tidal effects on all geodetic elements.

frqadjlovekhl.txt

```
doodson °/hr l l' F D Ω (δR δI)[2~72,e-5;73~120,e-4] H(e-5m)
  245655  28.43973  1  0  2  0  2       2      0   12099
  255555  28.98410  0  0  2  0  2       2      0   63192
  125755  12.85429  2  0  2  0  2     -29      3    -664
  127555  12.92714  0  0  2  2  2     -30      3    -802
  135645  13.39645  1  0  2  0  1     -45      5    -947
  135655  13.39866  1  0  2  0  2     -46      5   -5020
  137455  13.47151 -1  0  2  2  2     -49      5    -954
  145545  13.94083  0  0  2  0  1     -82      7   -4946
  145555  13.94303  0  0  2  0  2     -83      7  -26221
  147555  14.02517  0  0  0  2  0     -91      9     343
  153655  14.41456  1  0  2 -2  2    -168     14     194
  155445  14.48520 -1  0  2  0  1    -193     16     137
  155455  14.48741 -1  0  2  0  2    -194     16     741
  155655  14.49669  1  0  0  0  0    -197     16    2062
  155665  14.49890  1  0  0  0  1    -198     16     414
  157455  14.56955 -1  0  0  2  0    -231     18     394
  157465  14.57176 -1  0  0  2  1    -233     18      87
```

7.4.9 The Desai Ocean Polar Tide Coefficient File

The ocean polar tide arises from the centrifugal potential induced by polar motion acting upon the oceans. ETideLoad4.5 implements the self-consistent equilibrium model proposed

by Desai (2002), which accounts for continental boundaries, mass conservation, self-gravitation, and ocean floor loading.

The perturbations to the normalized geopotential coefficients are calculated as a function of polar shift parameters (m_1, m_2) using Equation (6.23) of the IERS Conventions (2010). The corresponding 360-degree coefficient file, desaiscopolecoef.txt, is stored in the iers folder.

ECMWF2006.dat | FES2004S1.dat | IERSeopc04.dat | GCN_L1_L2_30d_CF-CM.txt | desaiscopolecoef.txt

	n	m	Anm(Real)	Bnm(Real)	Anm(Imaginary)	Bnm(Imaginary)
2	1	0	1.8736759805448e-02	0.0000000000000e+00	2.9688884960424e-02	0.0000000000000e+00
3	1	1	2.8258913146935e-02	2.1774643075236e-02	2.3898264393684e-02	5.6771602236635e-02
4	2	0	-3.9555099024374e-03	0.0000000000000e+00	6.8390464271953e-04	0.0000000000000e+00
5	2	1	-2.4325330521304e-01	5.4680741193318e-03	5.4680741193318e-03	-1.9252111185300e-01
6	2	2	1.9102047023374e-02	1.1158297399424e-02	-1.5123770169928e-02	-2.4857839911518e-04
7	3	0	-2.0869478248378e-02	0.0000000000000e+00	-1.0775272844125e-02	0.0000000000000e+00
8	3	1	3.0809252024501e-02	7.4552838003486e-03	5.5937937407386e-03	6.6496877724041e-02
9	3	2	2.3295703062692e-02	3.7984356463618e-02	-2.1678456242839e-03	1.1232359168959e-02
10	3	3	7.9776020803848e-03	1.2502542787182e-02	-2.2341399966187e-02	-2.2979590161975e-02
11	4	0	-1.0612668622736e-02	0.0000000000000e+00	-1.5569196271270e-02	0.0000000000000e+00
12	4	1	1.3606306893006e-04	2.2051992576636e-03	2.0130037501025e-03	1.6323514549038e-02
13	4	2	1.1139374002795e-02	1.7031544962514e-02	-7.9621127289889e-03	-8.4440848505132e-04
14	4	3	-1.6100794768731e-02	1.4681986705593e-02	9.5178410813713e-03	-2.1017136590507e-02
15	4	4	4.3132021252707e-03	-4.6836271624465e-03	-2.9309550249205e-03	1.3175690530653e-02
16	5	0	7.0731357453056e-03	0.0000000000000e+00	-1.8023029843730e-03	0.0000000000000e+00
17	5	1	2.5644907587134e-03	-1.0076857169607e-02	-9.6273922883022e-03	-1.1684145258283e-02
18	5	2	-7.9615162895536e-03	2.0820461332209e-03	-3.0274671879191e-03	-1.0475800274156e-02
19	5	3	-1.1818705609675e-02	1.2063416189422e-02	-1.6584597520384e-02	-2.8253596831795e-02
20	5	4	9.2731253376468e-03	1.8353138561674e-02	-1.0870088052722e-02	4.7120935900411e-03
21	5	5	1.4460712839068e-02	-8.5510747244577e-03	8.9167437380844e-04	1.6048852898081e-02
22	6	0	7.4439256593180e-03	0.0000000000000e+00	-1.0670986469176e-03	0.0000000000000e+00
23	6	1	1.8261459881891e-02	-3.7775168887123e-03	-3.6768761254667e-03	-1.4329108864964e-03
24	6	2	-8.4568708595335e-03	2.5640802224787e-03	8.0976103423504e-03	-6.3983905389798e-03
25	6	3	-1.5355186088842e-02	1.8642889355748e-03	-9.6956523287846e-03	-2.2353328754893e-02
26	6	4	1.4142224508565e-03	-2.2076728030274e-03	-6.1060835758971e-03	1.4301205310949e-02
27	6	5	3.7744391579465e-03	1.6205935938625e-02	-7.4210466275681e-03	-2.8879881476777e-03
28	6	6	3.2420227193323e-03	-1.0204123402364e-03	6.5738366845630e-03	-6.6744309720085e-03

7.4.10 First-Degree Ocean Tidal Load Spherical Harmonic Coefficient File

The file OtideOne.dat is utilized to forecast the effects of ocean tidal loads on the Earth's Center of Mass (COM), or conversely, the effects of COM variations on geodetic elements due to ocean tides.

Generated by the function [Spherical Harmonic Analysis on Ocean Tidal Constituent Harmonic Constants], this file contains the degree-1 coefficients derived from the 34 tidal constituents of the FES2014b model.

Header: Includes the geocentric gravitational constant GM (×10¹⁴ m³/s²) and the Earth's equatorial radius a (m).

Records: Starting from the third row, each record lists the tidal constituent name, Doodson constant, the degree-1 order-0 spherical harmonic coefficient C_{10}^{+} (from in-phase amplitude), C_{10}^{-} (from out-of-phase amplitude), the degree-1 order-1 spherical harmonic coefficient C_{11}^{+} (from in-phase amplitude), C_{11}^{-} (from out-of-phase amplitude), and the degree-1 order-1 spherical harmonic coefficient S_{11}^{+} (from in-phase amplitude), S_{11}^{-} (from out-of-phase amplitude).

frqadjlovekhl.txt | OtideOne.dat

```
 3.986004418   6378137.00
name Doodson        C10+            C10-            C11+            C11-            S11+            S11-
 2N2 247.455  0.14379190E-08  0.72446933E-09  0.45556662E-09  0.50261431E-09  0.98234968E-09  0.28806626E-08
  J1 175.455  0.22805765E-08 -0.14599680E-07  0.11146859E-07  0.31354016E-08  0.49073923E-08  0.50239288E-08
  K1 165.555  0.65903198E-07 -0.23618735E-06  0.15240517E-06  0.54510351E-07  0.57951321E-07  0.91115166E-07
  K2 275.555  0.58820344E-08  0.78223673E-09  0.82634785E-08  0.17098158E-07  0.28274727E-08  0.95641986E-09
  L2 265.455  0.99527541E-09  0.43369491E-10  0.27208849E-08  0.18838893E-08 -0.93316186E-09 -0.31242492E-09
  M2 255.555  0.64086749E-07  0.33741274E-07  0.82092113E-07  0.76976307E-08 -0.39331272E-07  0.74234937E-07
  M3 355.555  0.51159035E-10  0.26216133E-10  0.20622631E-10 -0.16737336E-10 -0.74054752E-10 -0.32502465E-10
  M4 455.555 -0.12877739E-09 -0.82078020E-09  0.21241775E-09  0.89312487E-09 -0.11238411E-09 -0.11882183E-08
  M6 655.555  0.18174228E-08  0.30921490E-09  0.36600543E-09  0.36841599E-09 -0.72147727E-09 -0.13743491E-09
  M8 855.555 -0.59854172E-10 -0.29503418E-11  0.41858427E-10  0.58809710E-10 -0.34465624E-10  0.81925459E-11
  Mf  75.555  0.23994538E-07  0.23160661E-08  0.14961765E-07 -0.19050356E-07  0.57231952E-08 -0.38155669E-08
  Mm  65.455  0.12211587E-07 -0.10619733E-08 -0.13680094E-08 -0.93454574E-08  0.34149364E-08 -0.61740212E-09
  N2 245.655  0.16604395E-07  0.24692742E-08  0.10060051E-07  0.75631673E-09 -0.49125733E-09  0.20845840E-07
  N4 435.755 -0.11170849E-09 -0.41029169E-10  0.37178942E-10 -0.10703469E-09 -0.53442667E-10 -0.19926918E-10
  O1 145.555  0.23239277E-07 -0.16830188E-06  0.86481239E-07  0.11802879E-07  0.58555768E-07  0.34726677E-07
  P1 163.555  0.16600812E-07 -0.74602430E-07  0.48235157E-07  0.14146460E-07  0.16888410E-07  0.27904988E-07
  Q1 135.655  0.40244812E-08 -0.29117940E-07  0.15908436E-07  0.77164577E-09  0.12770867E-07  0.14909422E-08
  R2 274.554  0.21029138E-09  0.96276767E-10  0.25728894E-09  0.46084038E-09  0.53716115E-10  0.14500876E-09
  S1 164.556 -0.40129653E-08  0.48653114E-08 -0.48716881E-08  0.11419251E-07  0.74509139E-08 -0.34899535E-09
  S2 273.555  0.22430236E-07  0.94564697E-08  0.30377828E-07  0.49157638E-07 -0.61338730E-08  0.76805145E-08
  S4 491.555  0.32089047E-09  0.14407638E-09  0.12925319E-11  0.14038268E-09  0.10308541E-09  0.11742749E-09
  Sa  56.554  0.21793187E-09  0.12972260E-09  0.71714382E-10  0.49927099E-10 -0.42733149E-10 -0.53422994E-10
  T2 272.556  0.13719484E-08  0.73425584E-09  0.20944307E-08  0.29614380E-08  0.13767437E-09  0.10318216E-08
 MN4 445.655 -0.70793273E-09 -0.76823301E-10  0.24279253E-09 -0.66374018E-09 -0.14062685E-09  0.16716883E-09
 MS4 473.555  0.32582237E-09 -0.10684852E-08  0.10873236E-08  0.38092589E-09 -0.40703836E-09 -0.28009461E-09
 Msf  73.555  0.52032006E-09  0.12958178E-08  0.20898774E-09  0.69234415E-09  0.16108594E-08  0.36734674E-09
 Mtm  85.455  0.38057222E-08  0.89028662E-09  0.47545363E-08 -0.16109463E-08  0.13034435E-08  0.46197838E-10
 mu2 237.555  0.27230195E-08 -0.54548861E-09  0.80856645E-09  0.28475772E-08  0.30945151E-08  0.39961507E-08
 nu2 245.655  0.31512988E-08  0.13274377E-08  0.16643629E-08  0.77176190E-09 -0.34369557E-09  0.49489633E-08
 Ssa  57.555  0.85592993E-08 -0.21041028E-09 -0.85777470E-08 -0.10849053E-08  0.38854237E-09 -0.73333943E-09
eps2 227.655  0.15232320E-08 -0.54284574E-09  0.18709319E-08 -0.17678032E-09  0.14037532E-08 -0.64291979E-09
lam2 263.655  0.77975910E-09 -0.46145888E-09  0.29230225E-08 -0.81098933E-09 -0.68691816E-09 -0.10714953E-08
MKS2 257.555 -0.76338045E-11 -0.81694611E-10  0.81955321E-10  0.53313693E-09  0.52931064E-09  0.23733568E-09
Msqm  93.555  0.17382639E-09 -0.21085098E-11  0.98864729E-10  0.18391545E-09 -0.15315104E-09 -0.66456652E-11
```

7.4.11 First-Degree Atmosphere Tidal Load Spherical Harmonic Coefficient File

The file AirtdOne.dat is utilized to forecast the effects of surface atmospheric tidal loads on the Earth's Center of Mass (COM), or the effects of COM variations on geodetic elements due to atmospheric tides.

Generated by the function [Spherical Harmonic Analysis on Surface Atmosphere Tidal Harmonic Constants], this file contains the degree-1 coefficients derived from the 4 tidal constituents of the ECMWF-DCDA2006 model.

Header: Includes the geocentric gravitational constant GM ($\times 10^{14}$ m³/s²) and the Earth's equatorial radius a (m).

Records: Starting from the third row, each record lists the tidal constituent name, Doodson constant, the degree-1 order-0 spherical harmonic coefficient C_{10}^{+} (from in-phase amplitude), C_{10}^{-} (from out-of-phase amplitude), the degree-1 order-1 spherical harmonic coefficient C_{11}^{+} (from in-phase amplitude), C_{11}^{-} (from out-of-phase amplitude), and the degree-1 order-1 spherical harmonic coefficient S_{11}^{+} (from in-phase amplitude), S_{11}^{-} (from out-of-phase amplitude).

frqadjlovekhl.txt | OtideOne.dat | AirtdOne.dat

```
 3.986004418   6378137.00
name Doodson        C10+            C10-            C11+            C11-            S11+            S11-
  S1 164.556 -0.32743062E-08  0.17656303E-08 -0.73961700E-08 -0.43747341E-07 -0.53408107E-07 -0.10739535E-08
  S2 273.555 -0.63038465E-09  0.13745086E-08  0.76856572E-10  0.52369957E-08  0.33909106E-08 -0.10867279E-08
  Sa  56.565  0.82104410E-07 -0.16159641E-06 -0.35242403E-07 -0.82907617E-08  0.35038132E-07 -0.12165492E-06
 Ssa  57.555  0.65276776E-08  0.64836812E-07 -0.35849594E-07 -0.25040842E-07  0.12768923E-07  0.24915570E-07
```